



5 Inference on Model Parameters

Assume $\{Y_t\} \sim \text{ARMA}(p, q)$ being stationary with $E(Y_t) = \mu$, i.e.

$$\phi(B)(Y_t - \mu) = \theta(B)Z_t,$$

where $p = \deg(\phi)$ and $q = \deg(\theta)$. We will discuss the following.

1. Inference of $(\phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q)$ and μ given the order p and q .
2. Estimation of order p and q . (Included in the next tutorial)

5.1 Method of Moment Estimators

Method of Moment Estimators (MM)

Suppose $(X_1, \dots, X_n) \sim f_\theta$, and for all $i = 1, \dots, n$, $E\{g(X_i)\} = T(\theta)$ for some function g and T . Under regularity condition,

$$\overline{g(X)} = \frac{1}{n} \sum_{i=1}^n g(X_i) \xrightarrow{\text{pr}} E\{g(X_1)\} = T(\theta).$$

The method of moment suggest to **equate sample and population quantities** to solve for the estimator $\hat{\theta}_{\text{MM}} = \hat{\theta}_{\text{MM}}(X_1, \dots, X_n)$, i.e. to solve

$$T(\hat{\theta}_{\text{MM}}) = \frac{1}{n} \sum_{i=1}^n g(X_i)$$

❗ **Remark 1.** *There are several remarks on the method of moment.*

1. For $\theta \in \mathbb{R}^p$, we need at least p linearly dependent equation to estimate θ , i.e., assume $E\{g_k(X_i)\} = T_k(\theta)$ for $k = 1, \dots, p$ and solve $T_k(\hat{\theta}) = n^{-1} \sum_{i=1}^n g_k(X_i)$ for $k = 1, \dots, p$.
2. The method of moment estimator is not unique; see exercise 1 and 2. They might be very close to each other in value, but are still distinct, because
 - **(Theoretically Biased Building Blocks)** $E(C_k) \neq \gamma(k)$ and $E(r_k) \neq \rho(k)$ in general.
 - **(Lack of "Infinity Sample")** $C_k \xrightarrow{\text{pr}} \gamma(k)$ and $r_k \xrightarrow{\text{pr}} \rho(k)$ require " $n \rightarrow \infty$ "!

Assume $\{Y_t\} \sim \text{ARMA}(p, q)$ being stationary with ACVF $\gamma(\cdot)$ and ACF $\rho(\cdot)$. The following are some common building blocks for the construction of MM-estimators in the context of time series.

Population Quantity	Estimator
μ	$\bar{Y}_n = \frac{1}{n} \sum_{t=1}^n Y_t$
$\gamma(k)$	$C_k = \frac{1}{n} \sum_{t=1}^{n-k} (Y_t - \bar{Y})(Y_{t+k} - \bar{Y})$
$\rho(k)$	$r_k = \frac{C_k}{C_0}$

(★★☆) MM Estimator - Application in Classical Setting

 **Exercise 1.** Assume $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Po}(\mu)$.

- (a) By considering $E(X_1) = \mu$, suggest a MM-estimator $\hat{\mu}_{\text{MM}}^{(1)}$ of μ .
- (b) By considering $\text{Var}(X_1) = \mu$, suggest a MM-estimator $\hat{\mu}_{\text{MM}}^{(2)}$ of μ .

Attempt

(★★☆) MM Estimator - Application in Time Series Setting

 **Exercise 2.** Let $Y_t = Z_t + \phi Z_{t-1} + 2\phi Z_{t-2}$, where $Z_t \sim \text{WN}(0, \sigma^2)$.

- (a) Is $\{Y_t\}$ stationary? If so, evaluate ACVF $\gamma(\cdot)$ in terms of ϕ and σ^2 .
- (b) Suggest two approaches to construct the MM-estimator for ϕ and σ^2 .

Attempt

Recall for $\{Y_t\} \sim \text{AR}(p)$, we would take $\text{Cov}(\cdot, Y_{t-k})$ to obtain a system of linear equation in $\gamma(k)$'s.

- **(Previous Task)** Given $\phi_1, \dots, \phi_p$, solve for ACVF $\{\gamma(\cdot)\}$.
- **(Current Task)** Given the sample ACVF $C_1, \dots, C_p$, estimate $\hat{\phi}_1, \dots, \hat{\phi}_p$. (Not including σ^2)

We can simply replace (ϕ_i, γ_k) by $(\hat{\phi}_i, C_k)$ in the Yule-Walker equation to obtain the estimators $\hat{\phi}_1, \dots, \hat{\phi}_p$, and it is known as the **Yule-Walker Estimators**.

Yule-Walker Estimators (YW)

Definition 1. Assume $\{Y_t\} \sim \text{AR}(p)$ being causal, i.e. $Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + Z_t$, where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. The Yule-Walker Estimators $\hat{\phi}$ is defined as

$$\hat{\phi} = \begin{pmatrix} \hat{\phi}_1 \\ \vdots \\ \hat{\phi}_p \end{pmatrix} = \begin{pmatrix} 1 & r_1 & \cdots & r_{p-1} \\ r_1 & 1 & \cdots & r_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p-1} & r_{p-2} & \cdots & 1 \end{pmatrix}^{-1} \begin{pmatrix} r_1 \\ \vdots \\ r_p \end{pmatrix} = \begin{pmatrix} C_0 & C_1 & \cdots & C_{p-1} \\ C_1 & C_0 & \cdots & C_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{p-1} & C_{p-2} & \cdots & C_0 \end{pmatrix}^{-1} \begin{pmatrix} C_1 \\ \vdots \\ C_p \end{pmatrix}$$

❗ **Remark 2.** There are several remarks related to Yule-Walker Estimators

1. As by construction, Yule-Walker Estimators also belong to the class of MM-estimators.
2. The Yule-Walker Estimator does not provide an estimator of σ^2 . You can estimate σ^2 by

$$\text{Cov}(\cdot, Y_t) : \gamma(0) = \sum_{k=1}^p \phi_k \gamma(k) + \sigma^2 \quad \Rightarrow \quad \hat{\sigma}^2 = C_0 - \sum_{k=1}^p \hat{\phi}_k C_k.$$

3. You can also use the trick of taking $\text{Cov}(\cdot, Y_{t-k})$ to both sides of the MA(q) model to obtain the MM-estimators of $\theta_1, \dots, \theta_q$. However, $\text{Cov}(Y_t, Y_{t-k}) = \sum_{i=0}^{q-k} \theta_i \theta_{i+k}$ adopts a closed-form solution. So we would not say it to be an "equation" as there is nothing to be solved. The trick also works for solving MM-estimators in the ARMA model.

(☆☆) Yule-Walker Estimators

📎 **Exercise 3.** Assume $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t$, where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. Let $C_0 = 1$, $C_1 = 0.4$, and $C_2 = 0.25$.

- (a) Find the Yule-Walker estimators of $\phi = (\phi_1, \phi_2)^T$.
- (b) Suggest a method of moment estimator of σ^2 .

Attempt

5.2 Least Squares Type Estimators

5.2.1 Unconditional Least Squares (For AR Model)

Unconditional Least-Squares Method

Definition 2. Assume $\{Y_t\} \sim \text{AR}(p)$, i.e. $Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t$, where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. Let $Y_1, \dots, Y_n$ be the observed time series, define

$$\mathbf{X} = \begin{pmatrix} Y_p & Y_{p-1} & \cdots & Y_1 \\ Y_{p+1} & Y_p & \cdots & Y_2 \\ \vdots & \vdots & \ddots & \vdots \\ Y_{n-1} & Y_{n-2} & \cdots & Y_{n-p} \end{pmatrix} \quad \text{and} \quad \mathbf{Y} = \begin{pmatrix} Y_{p+1} \\ Y_{p+2} \\ \vdots \\ Y_n \end{pmatrix}.$$

Then the **unconditional least-squares estimators** of $\phi = (\phi_1, \dots, \phi_p)$ are given by

$$\bullet \hat{\phi} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \quad \bullet \hat{\sigma}^2 = (\mathbf{Y} - \mathbf{X} \hat{\phi})^T (\mathbf{Y} - \mathbf{X} \hat{\phi}) / (n - 2p).$$

❶ **Remark 3.** There are several remarks concerning unconditional least squares.

- The term "unconditional" arises because we do not need to initialize the noise sequence $\{Z_t\}$. Explicitly, we can write $Z_t = Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p}$ for $t = p+1, \dots, n$.
- As there is a total of $n - (p+1) + 1 = n - p$ estimated noise available and p predictors are used to build the regression, the denominator in $\hat{\sigma}^2$ is thus $(n - p) - p = n - 2p$.

Alternative Computation Method for ULS Estimators

It is indeed a regression problem. Hence, you can also find the ULS by the following procedure.

1. For $t = p+1, \dots, n$, write $Z_t = Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p}$.
2. Write $S(\phi_1, \dots, \phi_p) = \sum_{t=p+1}^n Z_t^2$ and let $(\hat{\phi}_1, \dots, \hat{\phi}_p) = \arg \min_{\phi_1, \dots, \phi_p} S(\phi_1, \dots, \phi_p)$.
3. Let $\hat{Z}_t = Y_t - \hat{\phi}_1 Y_{t-1} - \cdots - \hat{\phi}_p Y_{t-p}$ for $t = p+1, \dots, n$. Then $\hat{\sigma}^2 = (n - 2p)^{-1} \sum_{t=p+1}^n \hat{Z}_t^2$.

Asymptotic Normality of ULS

Theorem 1. Assume $\{Y_t\} \sim \text{AR}(p)$, i.e., $Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t$, where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. Let the ULS of $\phi = (\phi_1, \dots, \phi_p)^T$ be $\hat{\phi} = (\hat{\phi}_1, \dots, \hat{\phi}_p)^T$. Then as $n \rightarrow \infty$,

$$\sqrt{n}(\hat{\phi} - \phi) \xrightarrow{d} N_p(0, \sigma^2 \Gamma_p^{-1}), \quad \text{where} \quad \Gamma_p = \begin{pmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(p-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(p-1) & \gamma(p-2) & \cdots & \gamma(0) \end{pmatrix}$$

❶ **Remark 4.** There are several remarks on the inference of ULS.

- From Theorem 1, we know that for $\mathbf{w} = (w_1, \dots, w_p)^T \in \mathbb{R}^p$, as $n \rightarrow \infty$,

$$\sqrt{n}(\mathbf{w}^T \hat{\phi} - \mathbf{w}^T \phi) \xrightarrow{d} N(0, \sigma^2 \mathbf{w}^T \Gamma_p^{-1} \mathbf{w}).$$

It could be useful in testing a composite hypothesis, for example, $H_0: \phi_1 = \phi_2$.

- We can estimate Γ_p by $\hat{\Gamma}_p$ by replacing all $\gamma(k)$ by C_k .

(☆☆☆) Evaluation of ULS Estimators

 **Exercise 4.** Assume that $\{Y_t\} \sim \text{AR}(2)$, i.e., $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t$, where $Z_t \sim \text{WN}(0, \sigma^2)$.

- Let $Y_1 = -1$, $Y_2 = 1$, $Y_3 = 0$, $Y_4 = 4$, $Y_5 = -1$, $Y_6 = 3$.
- The parameters ϕ_1 , ϕ_2 and σ^2 are unknown.

- (a) Find the ULS estimator of ϕ_1 , ϕ_2 and σ^2 according to the definition 2.
- (b) Find the ULS estimator of ϕ_1 , ϕ_2 and σ^2 by directly minimizing the sum of squared error, i.e., by the alternative computation method.

Attempt

(★★☆) Inference on ULS Estimators

 **Exercise 5.** Let $\{Y_t\} \sim \text{AR}(2)$ with $n = 200$, $C_0 = 0.3$ and $C_1 = 0.2$. The unconditional least square estimators are given by $\hat{\phi}_1 = 0.5$, $\hat{\phi}_2 = 0.1$ and $\hat{\sigma}^2 = 0.25$,

- (a) Mathan claimed that an $\text{AR}(1)$ model should be used instead, do you agree with him?
- (b) Martin claimed that Y_{t-1} and Y_{t-2} have the same amount of impact on Y_t , do you agree with him?

Remark: Assume that we are conducting tests under the 95% confidence level.

Attempt

5.2.2 Conditional Least Squares (For MA/ARMA Model)

Recall that Y_t are **observable** for $t = 1, \dots, n$, while $\{Z_t\}$ is **unobservable**. For $\{Y_t\} \sim \text{AR}(p)$, we can write

$$Z_t = Y_t - \phi_1 Y_{t-1} - \dots - \phi_p Y_{t-p}$$

for $t = p+1, \dots, n$, which could be computed once the values of $\phi_1, \dots, \phi_p$ are known. In contrast, for $\{Y_t\} \sim \text{ARMA}(p, q)$, i.e.,

$$Z_t = Y_t - \phi_1 Y_{t-1} - \dots - \phi_p Y_{t-p} + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}.$$

As only $Y_1, \dots, Y_n$ are known to us, it is impossible for us to get a value of Z_t at any time t even though $\phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q$ are known. Therefore, we could NOT directly apply the least-square method. It motivates us to consider the **conditional-least square method**.

Conditional Least Squares (CLS) Method

Definition 3. Let $\{Y_t\} \sim \text{ARMA}(p, q)$ be **invertible**. Then

1. (**Initialization**) Assume $\tilde{Z}_s = Y_s = 0$ for all $s \leq 0$.
2. (**Sequential Estimation of Noise**) Let $\tilde{Z}_t = Y_t - \phi_1 Y_{t-1} - \dots - \phi_p Y_{t-p} + \theta_1 \tilde{Z}_{t-1} + \dots + \theta_q \tilde{Z}_{t-q}$.
3. (**Quantification of Error**) Define the sum of squared error as $S_*(\phi, \theta) = \sum_{t=1}^n \tilde{Z}_t^2$.

Then the **Conditional-Least Squares Estimators** are given by the minimizer of $S_*(\phi, \theta)$, i.e.

$$(\hat{\phi}, \hat{\theta}) = \arg \min_{\phi, \theta} S_*(\phi, \theta)$$

Remark 5. There are several remarks on the CLS method.

- The invertibility assumption is necessary for the following reasons.
 1. The CLS method assumed that Z_t can be estimated by past observations $\{Y_{t-k}\}_{k=1}^\infty$.
 2. Invertibility requires that the coefficient ψ_i satisfies $\sum_{k=0}^\infty |\psi_k| < \infty$, implying $\lim_{k \rightarrow \infty} \psi_k = 0$, i.e. the effect of the past observation on current noise would decay as the time lag increases. Hence, the error of assuming $\tilde{Z}_s = Y_s = 0$ for $s < 0$ would fade away as the sample size grows.
- There may not be an analytical/closed-form solution for the CLS estimator. Some common numerical methods, eg: the Newton-Raphson method or Expectation-Maximization (EM) algorithm, may help to give an estimated optimizer of $S_*(\phi, \theta)$.

(☆☆☆) Computation of CLS Estimator

Exercise 6. Assume $Y_t = Z_t - \theta Z_{t-1}$ to be invertible and $\{Z_t\} \sim \text{WN}(0, 1)$. Given $Y_1 = 0, Y_2 = 4, Y_3 = 5$, find the CLS estimator of θ .

Attempt

5.3 Maximum Likelihood Estimators

Let $X_1, \dots, X_n$ be random variables with joint-density $f_\theta(x_1, \dots, x_n)$ and likelihood L .

- $f_\theta(x_1, \dots, x_n)$: A measurement of the "probability" of the event " $X_1 = x_1, \dots, X_n = x_n$ " when the distribution is characterized by the parameter θ .
- $L(\theta|x_1, \dots, x_n)$: A measurement of "how likely" the true parameter would be θ when the observed data are given by $x_1, \dots, x_n$.

❗ Remark 6. In likelihood, we would NOT use the wording "probability of the parameter being θ " because θ is a fixed parameter without randomness (under frequentist setting).

Their value agrees, i.e. $f_\theta(x_1, \dots, x_n) = L(\theta|x_1, \dots, x_n)$, but differs in statistical meaning.

Maximum Likelihood Estimators

Definition 4. Let $(Y_1, \dots, Y_n)$ be time series observations with joint distribution $f_\theta(y_1, \dots, y_n)$, then the **Maximum Likelihood Estimator** of θ is defined as

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta|Y_1, \dots, Y_n)$$

In order to evaluate the MLE, we have to be able to write the joint density of a time series. There are two standard approaches to writing down the joint density.

Method I: Iterative Conditioning (More General)

Let $Y_1, \dots, Y_n$ be the observed time series data, then we can write the joint density as

$$f(Y_1, \dots, Y_n) = \left\{ \prod_{t=2}^n f(Y_t|Y_{t-1}, \dots, Y_1) \right\} f(Y_1)$$

❗ Remark 7. There are several remarks on Method I.

- It requires specification of marginal distribution of Y_1 , see Exercise 7.
- It provides flexibility for considering alternative noise other than normal.

(☆☆☆) Initialization of AR process

📎 Exercise 7. Let $Y_t = \phi Y_{t-1} + Z_t$ for $t \geq 2$ and $Y_1 = cZ_1$ for some c , where $Z_t \sim \text{WN}(0, \sigma^2)$. Suppose that there exists a constant c such that $\{Y_t\}_{t \geq 1}$ is weakly stationary, find value(s) of c .

Attempt

🔗 Takeaway 1. If we initialize a AR(1) process by $Y_1 = Z_1$, the process would be nonstationary. However, you should justify the existence of c formally by showing that $\text{Cov}(Y_t, Y_{t+k})$ is free of t .

(★★☆) MLE under AR(1) Model - Iterative Conditioning

 **Exercise 8.** Assume $Y_t = \phi Y_{t-1} + Z_t$ to be causal, where $\{Z_t\} \stackrel{iid}{\sim} N(0, \sigma^2)$.

- (a) Derive the log-likelihood function of $Y_1, \dots, Y_n$, i.e., $\ell(\phi, \sigma^2 | Y_1, \dots, Y_n) := \ln L(\phi, \sigma^2 | Y_1, \dots, Y_n)$.
- (b) Find the MLE of ϕ and σ^2 by iterative conditioning.
- (c) Given $Y_1 = 3$ and $Y_2 = 4$, find the value of $\hat{\phi}_{\text{MLE}}$ and $\hat{\sigma}_{\text{MLE}}^2$.

Attempt

Under regularity conditions, a linear combination of uncorrelated random variables follows normal distribution. We can model the joint distribution of causal time series by multivariate normal distribution.

Method II: Normality Assumption

Let $\{Y_t\} \sim \text{ARMA}(p, q)$ being causal and $\{Z_t\} \sim N(0, \sigma^2)$, $(Y_1, \dots, Y_n) \sim N_n(0, \Sigma)$, with density

$$f(y_1, \dots, y_n) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} \mathbf{y}^T \Sigma^{-1} \mathbf{y} \right\}, \text{ where } \Sigma = \begin{pmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(n-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(n-1) & \gamma(n-2) & \cdots & \gamma(0) \end{pmatrix}$$

and $\mathbf{y} = (y_1, \dots, y_n)$. In general, $\gamma(k)$ can be expressed in terms of σ^2 , ϕ 's and θ 's. Define

$$\left(\hat{\phi}_{\text{MLE}}, \hat{\theta}_{\text{MLE}}, \hat{\sigma}_{\text{MLE}}^2 \right) = \arg \max_{\phi, \theta, \sigma^2} L(\phi, \theta, \sigma^2 | Y_1, \dots, Y_n) = \arg \max_{\phi, \theta, \sigma^2} f(Y_1, \dots, Y_n)$$

Remark 8. Method II is simpler as it provides an explicit form of joint density, without the need for initialization of the process as in Exercise 7.

(★★☆) MLE under AR(1) model - Multivariate Normal Assumption

 **Exercise 9.** Assume $Y_t = \phi Y_{t-1} + Z_t$ to be causal, where $\{Z_t\} \sim N(0, \sigma^2)$.

- By noticing that (Y_1, Y_2) follows multivariate normal distribution, specify the joint-density of (Y_1, Y_2) .
- Given $Y_1 = 3$ and $Y_2 = 4$, find the log-likelihood of (Y_1, Y_2) .

Attempt

 **Takeaway 2.** You can solve $\frac{\partial \ell}{\partial \phi} = 0$ and $\frac{\partial \ell}{\partial \sigma^2} = 0$ for $\hat{\phi}_{\text{MLE}}$ and $\hat{\sigma}_{\text{MLE}}^2$. However, it does not have a closed-form solution. You may find the optimizer through numerical methods. Therefore, Method II provides an explicit form of joint density, but the evaluation of the estimator could be harder.

5.4 Summary of Estimation Methods

The following is a brief summary to the applicability of methods.

Method	AR(p) model	MA(q) or ARMA(p, q) model ($q \neq 0$)
Method of Moment (MM)	✓	✓
Yule-Walker (YW)	✓	✗
Unconditional Least-Squares (ULS)	✓	✗
Conditional Least-Squares (CLS)	✗	✓
Maximum Likelihood Estimator (MLE)	✓	✓

5.5 M-estimation

In general, most of the estimation methods mentioned above belong to the class of M -estimation.

M-estimator

Definition 5. Let $\theta \in \mathbb{R}^p$ be the estimand. Given the observations $(Y_1, \dots, Y_n)$. If there exists some function M such that the equation

$$M(\theta) = M(\theta, Y_{1:n}) = 0$$

has root $\hat{\theta}_M = \hat{\theta}_M(Y_1, \dots, Y_n)$, i.e., $M(\hat{\theta}_M) = 0$. Then, $\hat{\theta}_M$ is said to be a M estimator.

1. **Method of moment estimator:** Suppose that $E\{g_k(Y_1)\} = T_k(\theta)$ for $k = 1, \dots, p$, where $g_k(\cdot)$ and $T_k(\cdot)$ are univariate function. Then $\hat{\theta}_{MM}$ is the root to

$$M(\theta) = \begin{pmatrix} T_1(\theta) - \frac{1}{n} \sum_{t=1}^n g_1(Y_t) \\ \vdots \\ T_p(\theta) - \frac{1}{n} \sum_{t=1}^n g_p(Y_t) \end{pmatrix}.$$

2. **Unconditional least-square estimator:** $[\theta = (\phi_1, \dots, \phi_p)]$ For $\{Y_t\} \sim \text{AR}(p)$, write $Z_t(\phi_1, \dots, \phi_p) = Y_t - \sum_{k=1}^p \phi_k Y_{t-k}$ and $S_0(\theta) = \sum_{t=p+1}^n Z_t^2(\phi_1, \dots, \phi_p)$. Then $\hat{\theta}_{ULS}$ is the root to

$$M(\theta) = \nabla S_0(\theta).$$

3. **Conditional least-square estimator:** $[\theta = (\phi, \boldsymbol{\theta})]$ Let $S(\theta) = \sum_{t=1}^n \tilde{Z}_t^2(\phi, \boldsymbol{\theta})$. Then $\hat{\theta}_{CLS}$ is the root to

$$M(\theta) = \nabla S(\theta).$$

4. **Maximum likelihood estimator:** Let $L(\theta)$ be the likelihood function and assume it to be differentiable. Then $\hat{\theta}_{MLE}$ is the root to

$$M(\theta) = \nabla L(\theta).$$

Asymptotic normality of M -estimator

Theorem 2. Let θ_0 be the true value of the parameter and $\hat{\theta}$ as the M -estimator. Then

$$\sqrt{n}(\hat{\theta} - \theta_0) \approx \sqrt{n}\{M'(\theta_0)\}^{-1}M(\theta_0) \xrightarrow{d} N(0, \Sigma).$$

❗ **Remark 9.** M -estimator (in this course) is commonly referred to Z -estimator instead in the literature. The M -estimator (in the literature) have another statistical meaning.